

# SENSORS III

21M.370

IMUs, Encoders, and Photocells - oh my!

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# Lab 8

Assemble and test an IMU for in-class exploration

- motion sensing
- machine learning

Also includes info on assembling other sensors:

- lever switches
- encoders
- photocells
- hall effect sensors

# Performer Research

- What are their interests in terms of designing and performing with new instruments? What questions are they asking?
- How does their instrument work? What kinds of sensors, algorithms, software are they using?
- How does their performance practice incorporate their instrument?

## Groups:

- April 13 Chagall
- April 15 Jeff Snyder and Manta
- April 22a Mari Kimura
- April 22b Onyx Ashanti - Node 0

# CHAGALL



Ableton



Watch on

# JEFF SNYDER



Superbooth Berlin

Superbooth Berlin



Watch on

# MARI KIMURA



Mari Kimura



Watch on

# ONYX ASHANTI



Onyx Ashanti

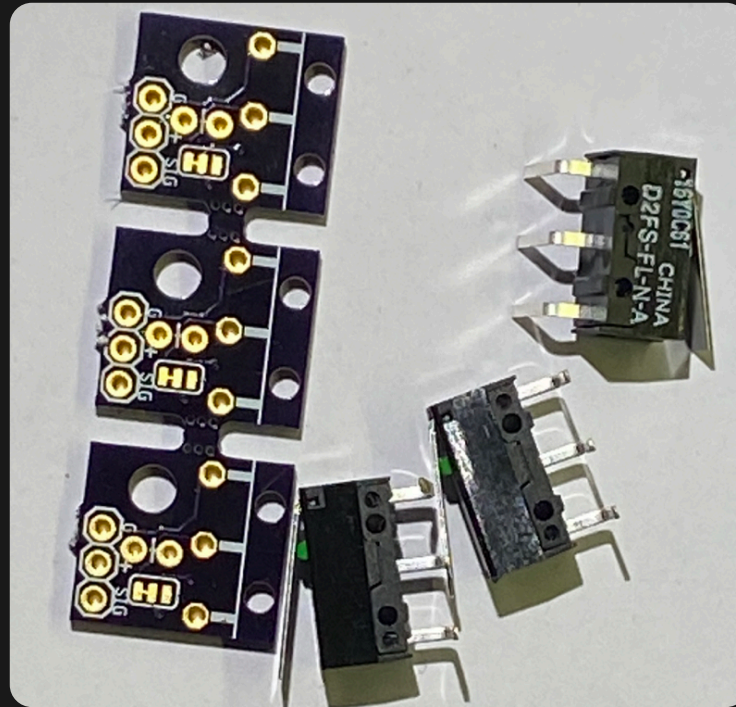


Watch on

# SENSORS



# limit switches

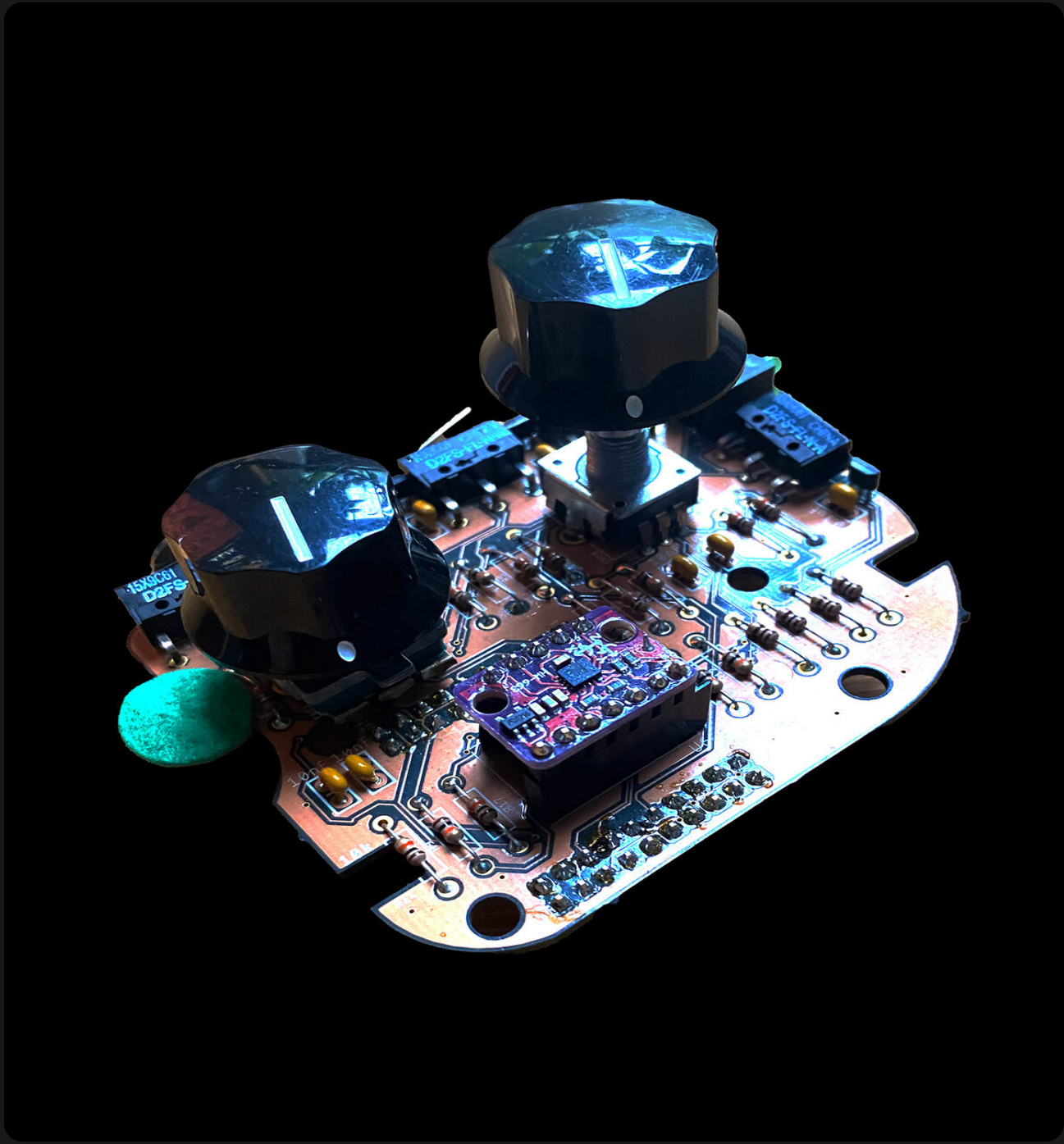


Extremely light action  
Comes in a variety of sizes

- we had a bunch of one size, with breakout PCBs for mounting
- great for finger switches!

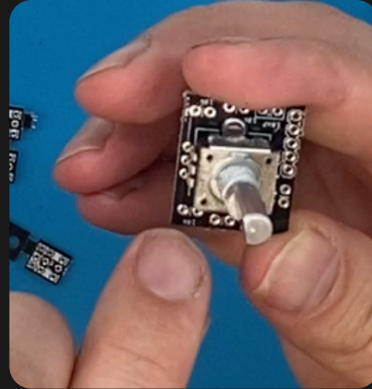
# limit switches

Chester!



Chester!

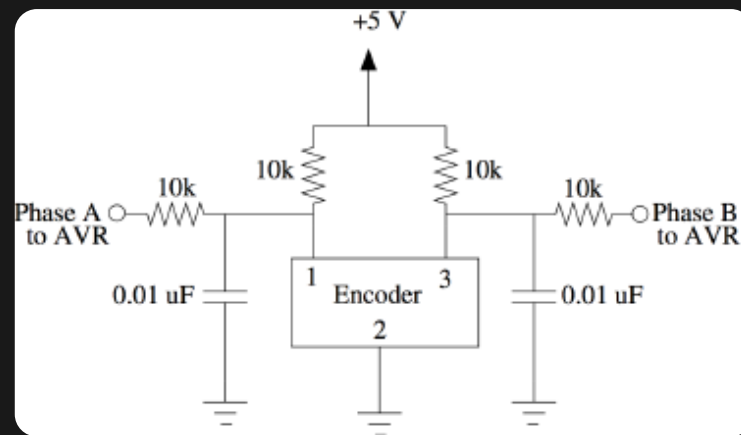
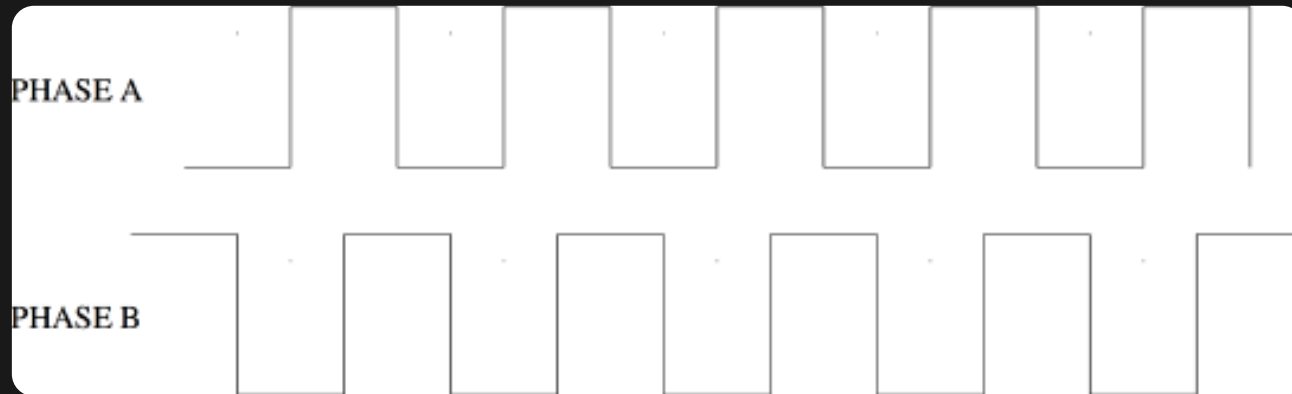
# rotary encoders



Measures rotational steps

- contains two switches with contacts with staggered phases
- interpretation of change of phases over time indicates direction of movement
- ‘quadrature encoding’
- switches will bounce just like all switches
- super important to limit bounces! best to do it before the ADC

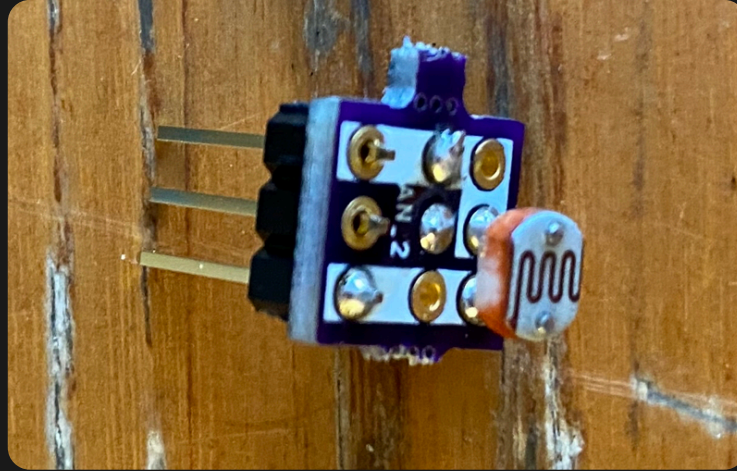
# encoders



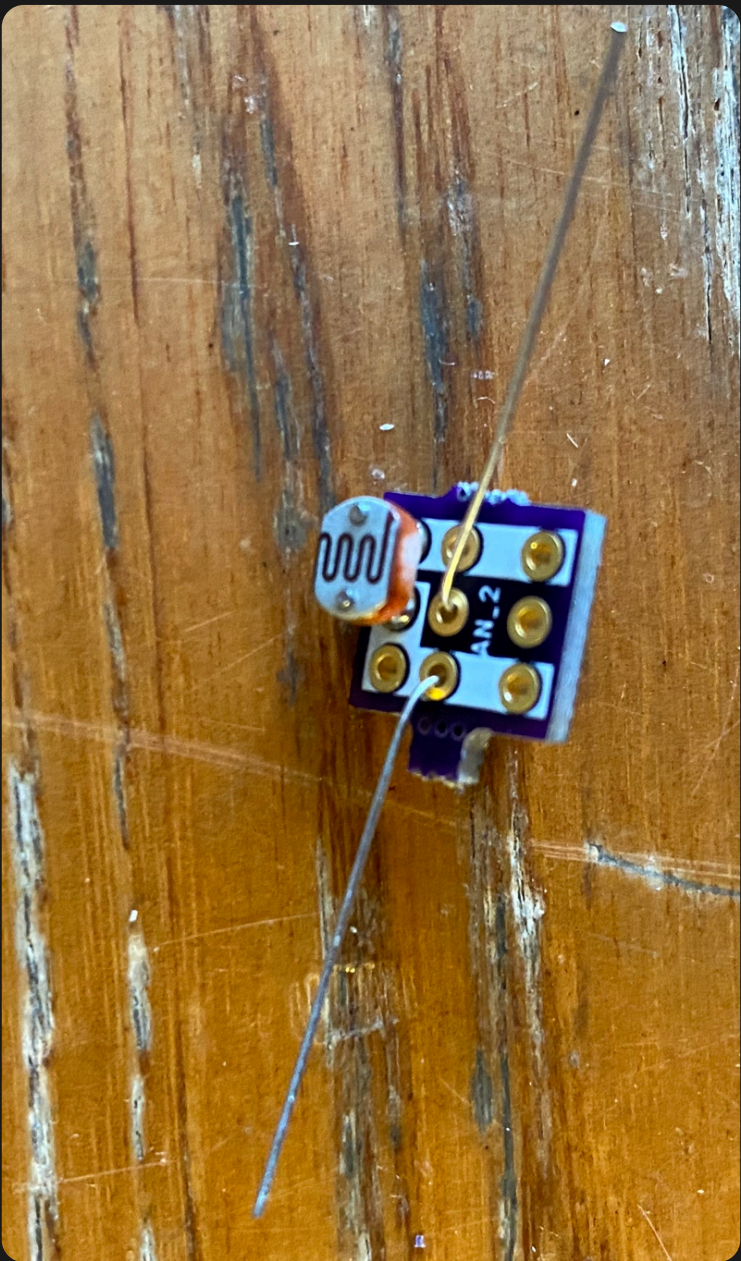
Source:

<https://ccrma.stanford.edu/workshops/mid2005/sensors/encoder.html>

# photocells







# hall effect switches





# Inertial Measurement Units

## Accelerometers

- Acceleration (not velocity)
- includes 1G of gravity!
- gravity is useful, allows for measurement of static tilt

## Gyroscopes

- rotation (not angle!)
- super responsive!

## Magnetometers

- direction to magnetic north
- slow, prone to interference

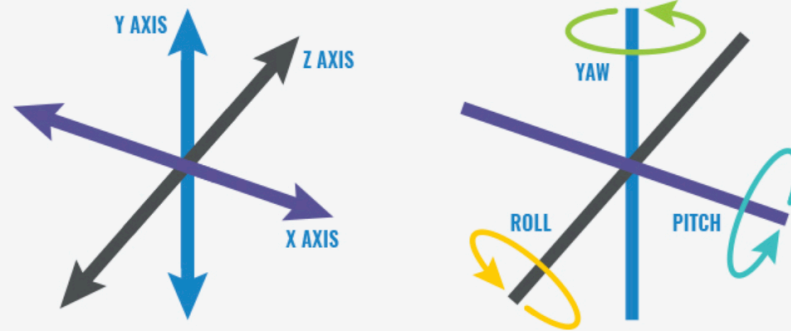
# Degrees of freedom

6/9 DoF sensors

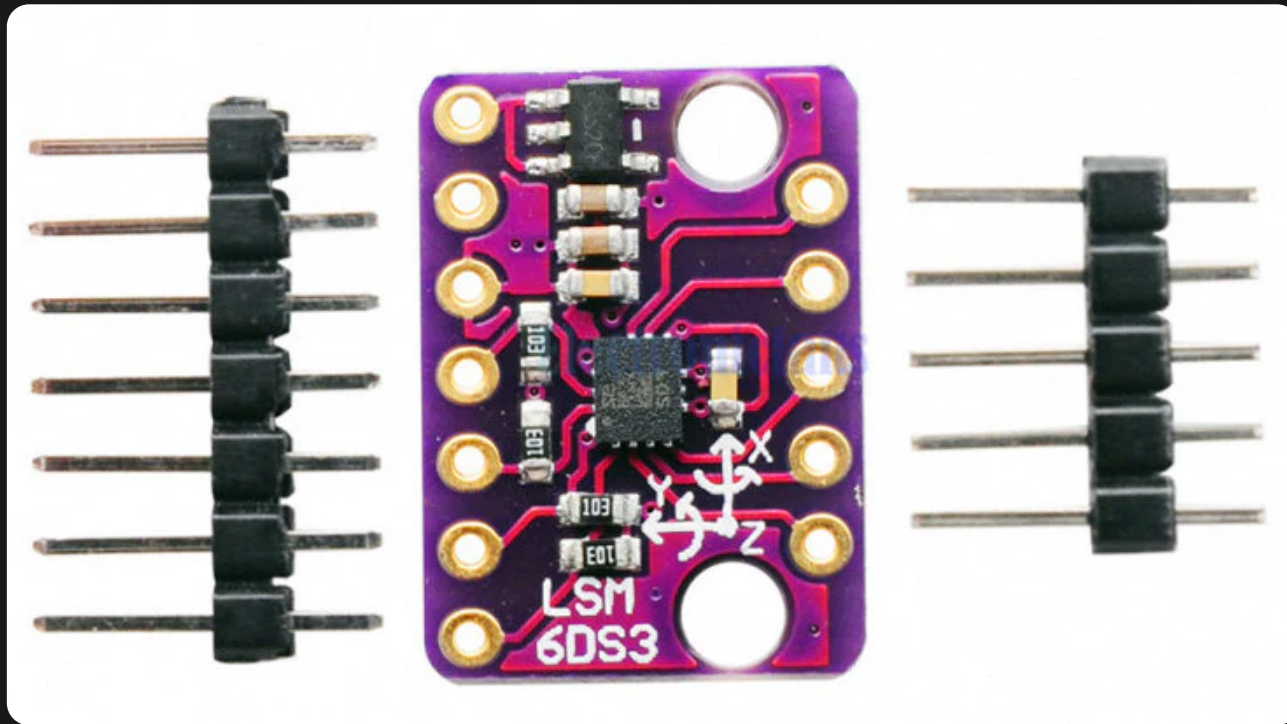
- degrees of freedom - independent measurement. XYZ = 3 degrees
- 6 DoF typically means XYZ accel and XYZ gyro
- 9 DoF includes magnetometer
- 9 DoF also sometimes referred to as MARG sensors (Magnetic, Angular Rate, and Gravity)

## SIX DEGREES OF FREEDOM

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# LSM6DS3



We will use a 6 DoF LSM6DS3 IMU.

- Uses I2C like MPR121
- Mounts directly to our PCB!

# IMU Gestures

what hand gestures are useful with an imu?

- tilt: pitch roll yaw (filter freq, timbre)
- shaking to inject energy (maracas, percussion)
- drum hit (in free air vs hitting a table)
- rotation
- signal - gesture recognition

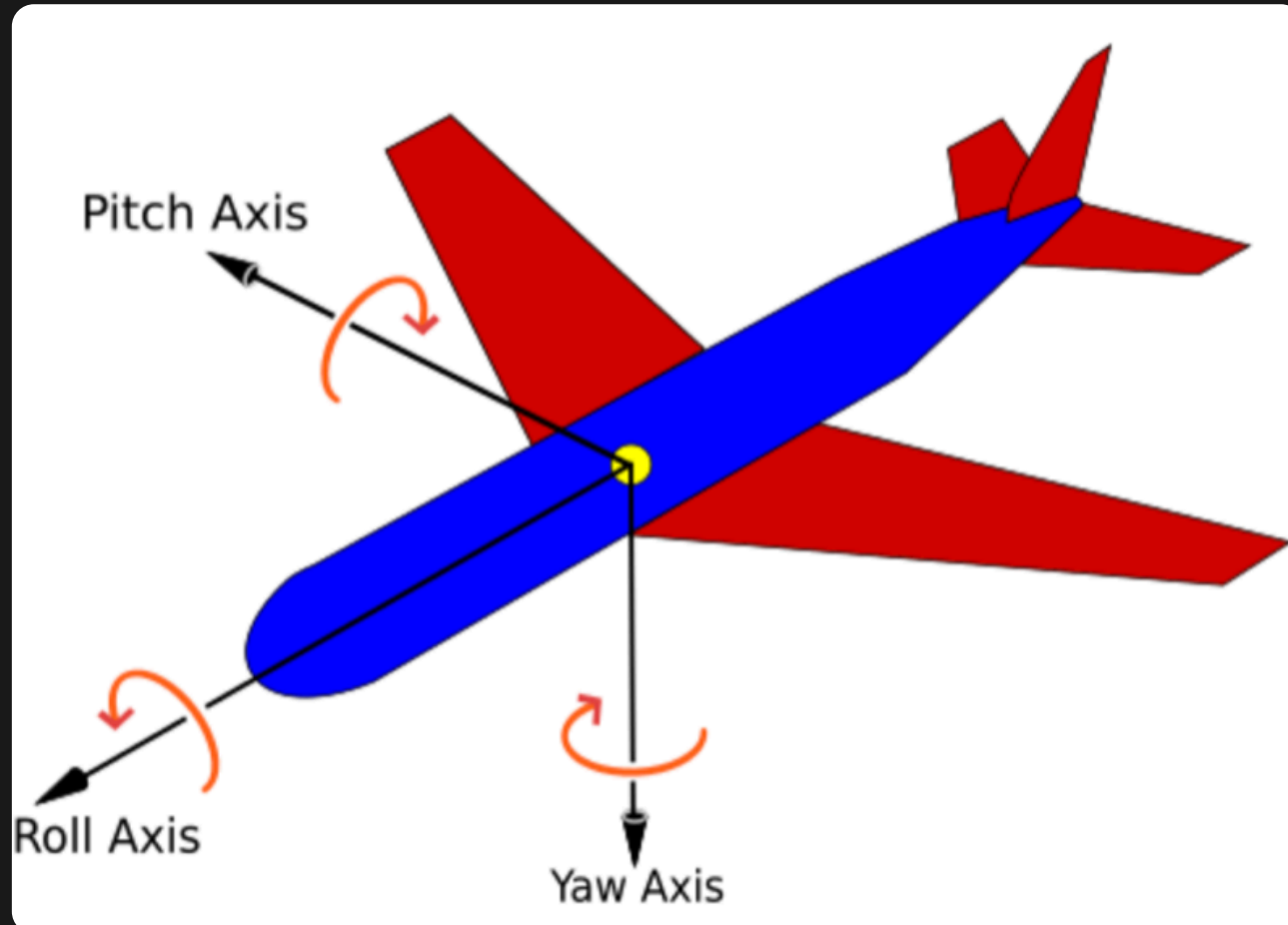
## Challenges

- velocity?
- slow, hard to segment fast passages
- absolute perception, perception in our sense is hard

# Tilt

Tilt is the most common usage of accelerometer data.

- It is easy to make sense of, physically
- It provides a static value which is easy to map
- Accelerometer will always register gravity
- gravity is a strong force!



# Tilt

Naive:  $x=\text{roll}$ ,  $y=\text{pitch}$

- axes are interactive and  $x$  acceleration doesn't map linearly to roll angle

The best way to calculate tilt is:

$$\text{pitch} = \arctan(-\text{accelX}, \sqrt{\text{accelY}^2 + \text{accelZ}^2})$$

$$\text{roll} = \arctan(\text{accelY}, \sqrt{\text{accelX}^2 + \text{accelZ}^2})$$

Yaw can't be accurately calculated using accelerometer - no change in static  $G$  force!

# Rotation

Absolute angle needs a fixed reference

- for pitch and roll, gravity provides the reference
- for gyro based yaw, we need to provide that reference explicitly  
Naive:  $\text{yaw} += \text{gyroZ}$
- will drift badly
- often a trigger is used to redefine 0 degrees yaw



# Triggers

‘Air Drums’

- at what point is a drum struck?
- not interested in specific angles, but overall acceleration  
Magnitude of acceleration: calculates total acceleration independent of axes
- $\text{magnitude} = \sqrt{x^2 + y^2 + z^2}$
- subtract the static gravity value

# Detecting a trigger

Set a threshold and trigger hit immediately

- instantaneous value past trigger might not be peak acceleration  
Set a threshold and trigger hit on first negative acceleration value after threshold is passed
- may add latency
- allows for accurate magnitude detection  
Always need some way to prevent retriggers

# Shaking

Overall running average of magnitude

- easy to calculate
- neatly correlates to physical activity

Calculation:

- lowpass filter
- leaky integrator: continuously integrate magnitude, add constant leak to bring value back to 0
- $\text{shake} = (\text{mag} + \text{shake} * \text{leak})$
- $\text{shake} = \text{mag} + \text{shake} - \text{leak}$